INTELLIGENT DECISION LAB

Towards Sample Efficient RL

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Bio

- **Postdoc Scholar @ CITI, Academia Sinica**
  - Reinforcement Learning
- **Postdoc @ IMADA, University of Southern Denmark**
  - Reinforcement Learning, Probabilistic Methods
- **Ph.D. @ DIKU, University of Copenhagen**
  - Statistical Learning Theory
- **Visiting Researcher @ CWI, Netherlands**
- **Research Assistant @ IIS, Academia Sinica**
  - Online Learning & Bandit & Lifelong Learning
- **Bachelor in Physics @ NTU**

Where Have You Heard About Reinforcement Learning?

DQN (2015)

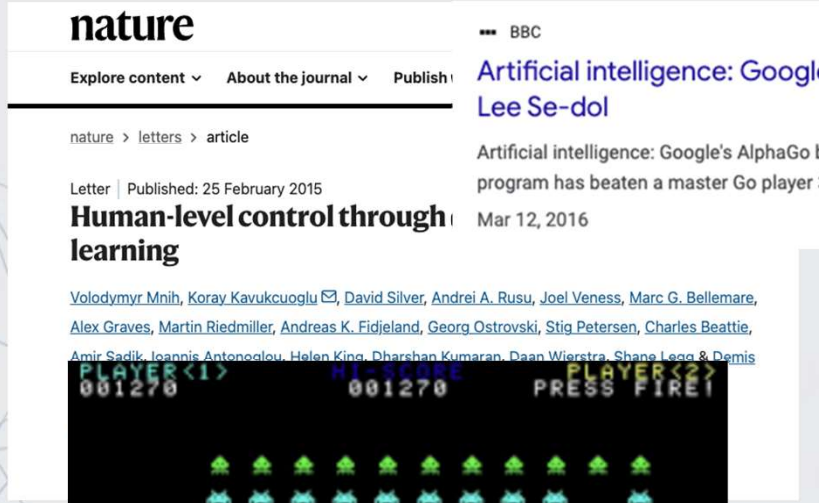
AlphaGo (2016)

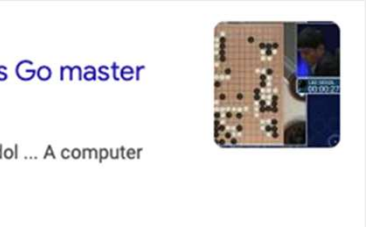
RLHF for LLMs (2022~)

Turing Award (2024)

Led the Development of

Cornerstone AI Technology

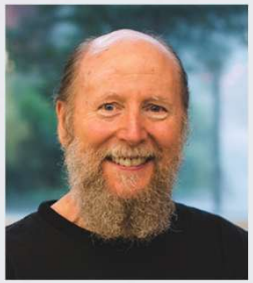




Claude

LLaMA

Gemini



Reinforcement Learning

A paradigm where an agent learns to make goal-directed decisions by interacting and using feedback signals from the environment to update itself, aiming to maximize long-term objectives.

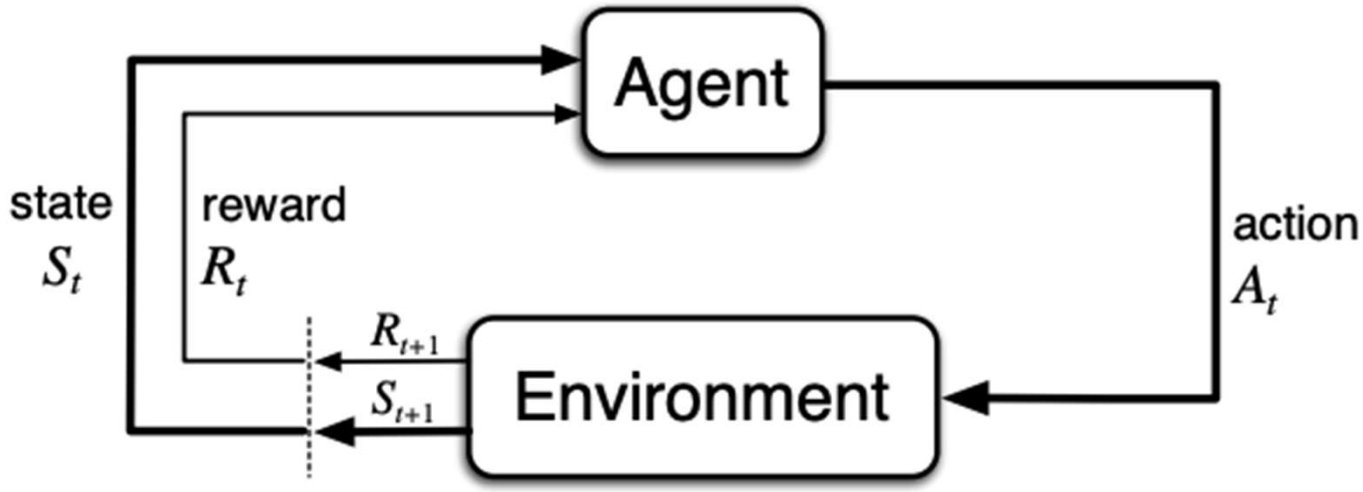
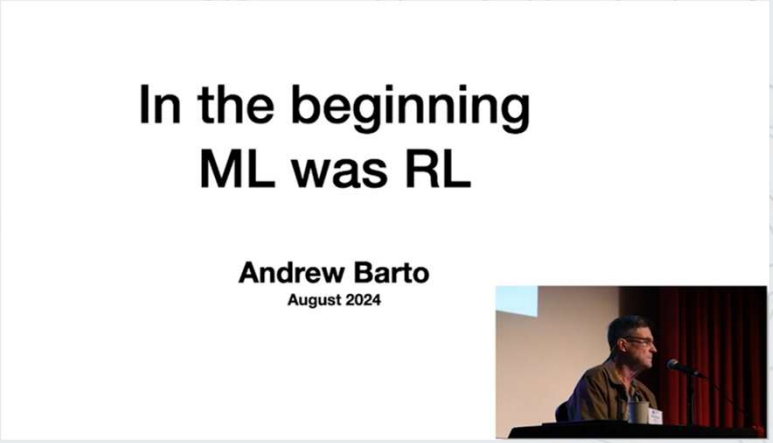


Figure 3.1: The agent–environment interaction in reinforcement learning.

Source: Sutton and Barto (2018)

(Concept of) RL Has a Long History



1911 — Law of Effect (Edward Thorndike)

1948 — Pleasure-pain system (Alan Turing)

1952 — Multi-armed bandit (Herbert Robbins)

1957 — Bellman Equation (Richard Bellman)

1972 — The "Hedonistic Neuron" Hypothesis (A. Harry Klopf)

1980s — TD Algorithm, Actor-Critic (Sutton & Barto)

Q Learning (Chris Watkins)

1996 — Neurodynamic programming

(Dimitri Bertsekas & John Tsitsiklis)

Law of Effect (Edward Thorndike, 1911)

"Responses generating a pleasing impact in a situation are more likely to occur again in the same situation, while responses that cause discomfort are less likely to occur again in the same situation."

Edward Thorndike (psychologist)

Image source: https://en.wikipedia.org/wiki/Edward_Thorndike

Klopf's Hypothesis (1972)

"If the action potential is followed within an appropriate time period by an increase in reward, the efficacies of all eligible synapses increase (or decrease in the case of punishment)"

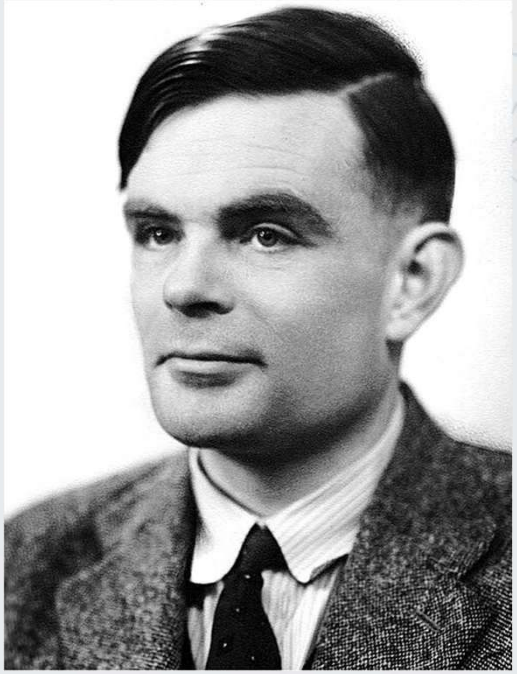
- From neuroscience perspective
- Inspire Barto and Sutton to develop TD learning

A. Harry Klopf
(computer scientist and electric engineer)

Image source: https://www.chessprogramming.org/A._Harry_Klopf

Pleasure-Pain System (Alan Turing, 1948)

"When a configuration is reached for which the action is undetermined, a random choice for the missing data is made and the appropriate entry is made in the description, tentatively, and is applied. When a pain stimulus occurs all tentative entries are cancelled, and when a pleasure stimulus occurs they are all made permanent."



Alan Turing
(computer scientist)

Image source: https://en.wikipedia.org/wiki/Alan_Turing

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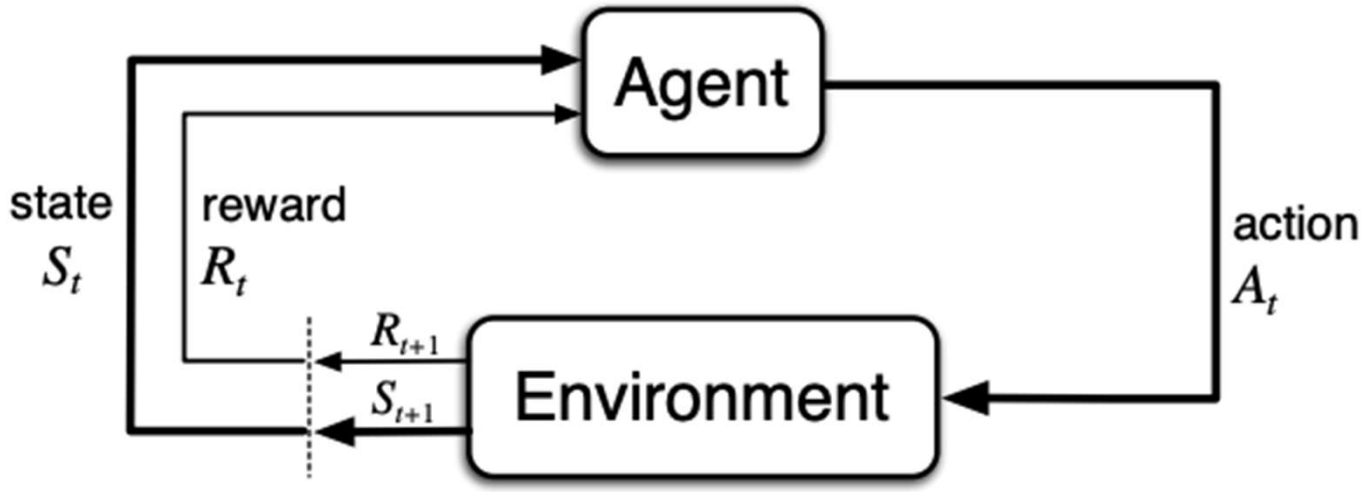

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Bellman Equation (Richard E. Bellman, 1956)

Principle of Optimality: the optimal solution to a dynamic optimization problem can be found by combining the optimal solutions to its sub-problems.

- From Optimal Control Theory
  - how to control dynamical systems over time that minimize a cost function
- No MDP (M=(S, A, R, P)) required
- MDP from Operations Research
  - Formal model of sequential decision-making under uncertainty

Richard E. Bellman (mathematician)

Multi-Armed Bandit

"In the long run, if a real preference exists between the two treatments, the expected saving by continued application of this method of apportionment rather than by making immediate final decision is sensibly 1 - P of individuals subsequently treated." (William R. Thompson, 1933)

"The whole problem lies in deciding how to draw the sample. ... In fact, the problem represents in a simplified way the general question of how we learn - or should learn - from past experience." (Herbert Robbins, 1952)

Regret of the process:

$$R_T = E\left[\sum_{t=1}^{T}(\mu^* - \mu_{I_t})\right]$$

(Lai and Robbins, 1985)

Reinforcement Learning

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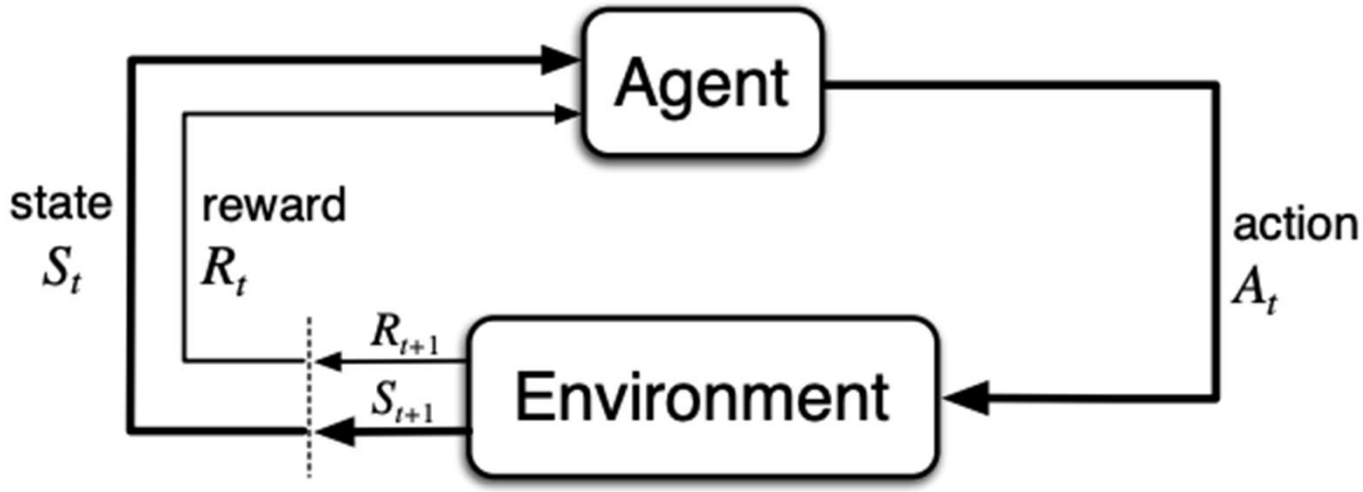

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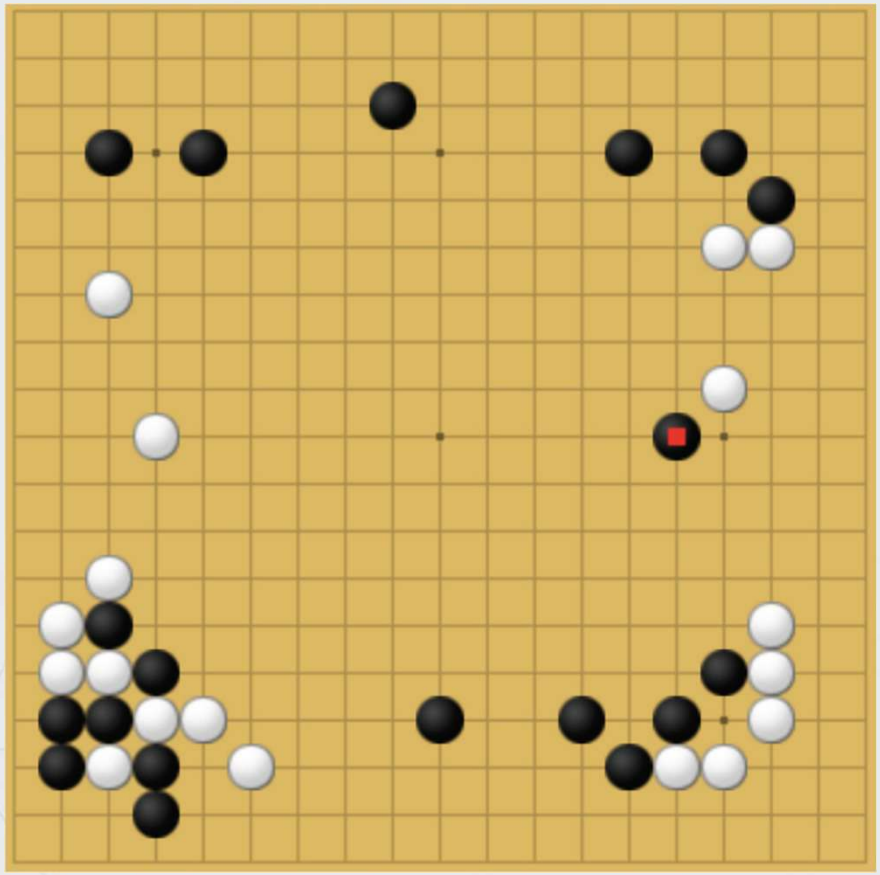
RL: The Path to Intelligence

Passive Data-Driven AI	Active RL
● **Input**: Vast, static datasets. ● Excellent at summarizing and mimicking patterns found in the given data.	● Actively **explores** to collect data relevant to the goal. ● **Discovers new solutions.**

Ultimately, the intelligent agent must have the ability to build new knowledge from trying things and seeing consequences.

This is what RL enables.

Example 1: AlphaGo Move 37



Source: https://www.researchgate.net

Example 2: AlphaTensor for Matrix Multiplication

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \times \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{bmatrix}$$

Volker Strassen (1969)

Standard algorithm	Strassen's algorithm
$h_1 = a_{1,1}\ b_{1,1}$	$h_1 = (a_{1,1} + a_{2,2})\ (b_{1,1} + b_{2,2})$
$h_2 = a_{1,1}\ b_{1,2}$	$h_2 = (a_{2,1} + a_{2,2})\ b_{1,1}$
$h_3 = a_{1,2}\ b_{2,1}$	$h_3 = a_{1,1}\ (b_{1,2} - b_{2,2})$
$h_4 = a_{1,2}\ b_{2,2}$	$h_4 = a_{2,2}\ (-b_{1,1} + b_{2,1})$
$h_5 = a_{2,1}\ b_{1,1}$	$h_5 = (a_{1,1} + a_{1,2})\ b_{2,2}$
$h_6 = a_{2,1}\ b_{1,2}$	$h_6 = (-a_{1,1} + a_{2,1})\ (b_{1,1} + b_{1,2})$
$h_7 = a_{2,2}\ b_{2,1}$	$h_7 = (a_{1,2} - a_{2,2})(b_{2,1} + b_{2,2})$
$h_8 = a_{2,2}\ b_{2,2}$	
$c_{1,1} = h_1 + h_3$	$c_{1,1} = h_1 + h_4 - h_5 + h_7$
$c_{1,2} = h_2 + h_4$	$c_{1,2} = h_3 + h_5$
$c_{2,1} = h_5 + h_7$	$c_{2,1} = h_2 + h_4$
$c_{2,2} = h_6 + h_8$	$c_{2,2} = h_1 - h_2 + h_3 + h_6$

Size (n, m, p)	Best method known	Best rank known	AlphaTensor rank Modular	AlphaTensor rank Standard
(2, 2, 2)	(Strassen, 1969)[2]	7	7	7
(3, 3, 3)	(Laderman, 1976)[15]	23	23	23
(4, 4, 4)	(Strassen, 1969)[2] (2, 2, 2) ⊗ (2, 2, 2)	49	47	49
(5, 5, 5)	(3, 5, 5) + (2, 5, 5)	98	96	98
(2, 2, 3)	(2, 2, 2) + (2, 2, 1)	11	11	11
(2, 2, 4)	(2, 2, 2) + (2, 2, 2)	14	14	14
(2, 2, 5)	(2, 2, 2) + (2, 2, 3)	18	18	18
(2, 3, 3)	(Hopcroft and Kerr, 1971)[16]	15	15	15
(2, 3, 4)	(Hopcroft and Kerr, 1971)[16]	20	20	20
(2, 3, 5)	(Hopcroft and Kerr, 1971)[16]	25	25	25
(2, 4, 4)	(Hopcroft and Kerr, 1971)[16]	26	26	26
(2, 4, 5)	(Hopcroft and Kerr, 1971)[16]	33	33	33
(2, 5, 5)	(Hopcroft and Kerr, 1971)[16]	40	40	40
(3, 3, 4)	(Smirnov, 2013)[18]	29	29	29
(3, 3, 5)	(Smirnov, 2013)[18]	36	36	36
(3, 4, 4)	(Smirnov, 2013)[18]	38	38	38
(3, 4, 5)	(Smirnov, 2013)[18]	48	47	47
(3, 5, 5)	(Sedoglavic and Smirnov, 2021)[19]	58	58	58
(4, 4, 5)	(4, 4, 2) + (4, 4, 3)	64	63	63
(4, 5, 5)	(2, 5, 5) ⊗ (2, 1, 1)	80	76	76

Sources: https://deepmind.google/blog/discovering-novel-algorithms-with-alphatensor/ ; https://www.nature.com/articles/s41586-022-05172-4

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$h_1 = a_{3,2}\left(-b_{2,1} - b_{2,5} - b_{3,1}\right)$

$h_2 = \left(a_{2,2} + a_{2,5} - a_{3,5}\right)\left(-b_{2,5} - b_{5,1}\right)$

$h_3 = \left(-a_{3,1} - a_{4,1} + a_{4,2}\right)\left(-b_{1,1} + b_{2,5}\right)$

$h_4 = \left(a_{1,2} + a_{1,4} + a_{3,4}\right)\left(-b_{2,5} - b_{4,1}\right)$

$h_5 = \left(a_{1,5} + a_{2,2} + a_{2,5}\right)\left(-b_{2,4} + b_{5,1}\right)$

$h_6 = \left(-a_{2,2} - a_{2,5} - a_{4,5}\right)\left(b_{2,3} + b_{5,1}\right)$

$h_7 = \left(-a_{1,1} + a_{4,1} - a_{4,2}\right)\left(b_{1,1} + b_{2,4}\right)$

$h_8 = \left(a_{3,2} - a_{3,3} - a_{4,3}\right)\left(-b_{2,3} + b_{3,1}\right)$

$h_9 = \left(-a_{1,2} - a_{1,4} + a_{4,4}\right)\left(b_{2,3} + b_{4,1}\right)$

$h_{10} = \left(a_{2,2} + a_{2,5}\right) b_{5,1}$

$h_{11} = \left(-a_{2,1} - a_{4,1} + a_{4,2}\right)\left(-b_{1,1} + b_{2,2}\right)$

$h_{12} = \left(a_{4,1} - a_{4,2}\right) b_{1,1}$

$h_{13} = \left(a_{1,2} + a_{1,4} + a_{2,4}\right)\left(b_{2,2} + b_{4,1}\right)$

$h_{14} = \left(a_{1,3} - a_{3,2} + a_{3,3}\right)\left(b_{2,4} + b_{3,1}\right)$

$h_{15} = \left(-a_{1,2} - a_{1,4}\right) b_{4,1}$

$h_{16} = \left(-a_{3,2} + a_{3,3}\right) b_{3,1}$

$h_{17} = \left(a_{1,2} + a_{1,4} - a_{2,1} + a_{2,2} - a_{2,3} + a_{2,4} - a_{3,2} + a_{3,3} - a_{4,1} + a_{4,2}\right) b_{2,2}$

$h_{18} = a_{2,1}\left(b_{1,1} + b_{1,2} + b_{5,2}\right)$

$h_{19} = -a_{2,3}\left(b_{3,1} + b_{3,2} + b_{5,2}\right)$

$h_{20} = \left(-a_{1,5} + a_{2,1} + a_{2,3} - a_{2,5}\right)\left(-b_{1,1} - b_{1,2} + b_{1,4} - b_{5,2}\right)$

$h_{21} = \left(a_{2,1} + a_{2,3} - a_{2,5}\right) b_{5,2}$

$h_{22} = \left(a_{1,3} - a_{1,4} - a_{2,4}\right)\left(b_{1,1} + b_{1,2} - b_{1,4} - b_{3,1} - b_{3,2} + b_{3,4} + b_{4,4}\right)$

$h_{23} = a_{1,3}\left(-b_{3,1} + b_{3,4} + b_{4,4}\right)$

$h_{24} = a_{1,5}\left(-b_{4,4} - b_{5,1} + b_{5,4}\right)$

$h_{25} = -a_{1,1}\left(b_{1,1} - b_{1,4}\right)$

$h_{26} = \left(-a_{1,3} + a_{1,4} + a_{1,5}\right) b_{4,4}$

$h_{27} = \left(a_{1,3} - a_{3,1} + a_{3,3}\right)\left(b_{1,1} - b_{1,4} + b_{1,5} + b_{3,5}\right)$

$h_{28} = -a_{3,4}\left(-b_{3,5} - b_{4,1} - b_{4,5}\right)$

$h_{29} = a_{3,1}\left(b_{1,1} + b_{1,5} + b_{3,5}\right)$

$h_{30} = \left(a_{3,1} - a_{3,3} + a_{3,4}\right) b_{3,5}$

$h_{31} = \left(-a_{1,4} - a_{1,5} - a_{3,4}\right)\left(-b_{4,4} - b_{5,1} + b_{5,4} - b_{5,5}\right)$

$h_{32} = \left(a_{2,1} + a_{4,1} + a_{4,4}\right)\left(b_{1,3} - b_{4,1} - b_{4,2} - b_{4,3}\right)$

$h_{33} = a_{4,3}\left(-b_{3,1} - b_{3,3}\right)$

$h_{34} = a_{4,4}\left(-b_{1,3} + b_{4,1} + b_{4,3}\right)$

$h_{35} = -a_{4,5}\left(b_{1,3} + b_{5,1} + b_{5,3}\right)$

$h_{36} = \left(a_{2,3} - a_{2,5} - a_{4,5}\right)\left(b_{3,1} + b_{3,2} + b_{3,3} + b_{5,2}\right)$

$h_{37} = \left(-a_{4,1} - a_{4,4} + a_{4,5}\right) b_{1,3}$

$h_{38} = \left(-a_{2,3} - a_{3,1} + a_{3,3} - a_{3,4}\right)\left(b_{3,5} + b_{4,1} + b_{4,2} + b_{4,5}\right)$

$h_{39} = \left(-a_{3,1} - a_{4,1} - a_{4,4} + a_{4,5}\right)\left(b_{1,3} + b_{5,1} + b_{5,3} + b_{5,5}\right)$

$h_{40} = \left(-a_{1,3} + a_{1,4} + a_{1,5} - a_{4,4}\right)\left(-b_{3,1} - b_{3,3} + b_{3,4} + b_{4,4}\right)$

$h_{41} = \left(-a_{1,1} + a_{4,1} - a_{4,5}\right)\left(b_{1,3} + b_{3,1} + b_{3,3} - b_{3,4} + b_{5,1} + b_{5,3} - b_{5,4}\right)$

$h_{42} = \left(-a_{2,1} + a_{2,5} - a_{3,5}\right)\left(-b_{1,1} - b_{1,2} - b_{1,5} + b_{4,1} + b_{4,2} + b_{4,5} - b_{5,2}\right)$

$h_{43} = a_{2,4}\left(b_{4,1} + b_{4,2}\right)$

$h_{44} = \left(a_{2,3} + a_{3,2} - a_{3,3}\right)\left(b_{2,2} - b_{3,1}\right)$

$h_{45} = \left(-a_{3,3} + a_{3,4} - a_{4,3}\right)\left(b_{3,5} + b_{4,1} + b_{4,3} + b_{4,5} + b_{5,1} + b_{5,3} + b_{5,5}\right)$

$h_{46} = -a_{3,5}\left(-b_{5,1} - b_{5,5}\right)$

$h_{47} = \left(a_{2,1} - a_{2,5} - a_{3,1} + a_{3,5}\right)\left(b_{1,1} + b_{1,2} + b_{1,5} - b_{4,1} - b_{4,2} - b_{4,5}\right)$

$h_{48} = \left(-a_{2,3} + a_{3,3}\right)\left(b_{2,2} + b_{3,2} + b_{3,5} + b_{4,1} + b_{4,2} + b_{4,5}\right)$

$h_{49} = \left(-a_{1,1} - a_{1,3} + a_{1,4} + a_{1,5} - a_{2,1} - a_{2,3} + a_{2,4} + a_{2,5}\right)\left(-b_{1,1} - b_{1,2} + b_{1,4}\right)$

$h_{50} = \left(-a_{1,4} - a_{2,4}\right)\left(b_{2,2} - b_{3,1} - b_{3,2} + b_{3,4} - b_{4,2} + b_{4,4}\right)$

$h_{51} = a_{2,2}\left(b_{2,1} + b_{2,2} - b_{5,1}\right)$

$h_{52} = a_{4,2}\left(b_{1,1} + b_{2,1} + b_{2,3}\right)$

$h_{53} = -a_{1,2}\left(-b_{2,1} + b_{2,4} + b_{4,1}\right)$

$h_{54} = \left(a_{1,2} + a_{1,4} - a_{2,2} - a_{2,5} - a_{3,2} + a_{3,3} - a_{4,2} + a_{4,3} - a_{4,4} - a_{4,5}\right) b_{2,3}$

$c_{1,1} = -h_{10} + h_{12} + h_{14} - h_{15} - h_{16} + h_{53} + h_5 - h_{66} - h_7$

$c_{2,1} = h_{10} + h_{11} - h_{12} + h_{13} + h_{15} + h_{16} - h_{17} - h_{44} + h_{51}$

$c_{3,1} = h_{10} - h_{12} + h_{15} + h_{16} - h_1 + h_2 + h_3 - h_4 + h_{75}$

$c_{4,1} = -h_{10} + h_{12} - h_{15} - h_{16} + h_{52} + h_{54} - h_6 - h_8 + h_9$

$c_{1,2} = h_{13} + h_{15} + h_{20} + h_{21} - h_{22} + h_{23} + h_{25} - h_{43} + h_{49} + h_{50}$

$c_{2,2} = -h_{11} + h_{12} - h_{13} - h_{15} - h_{16} + h_{17} + h_{18} - h_{19} - h_{21} + h_{43} + h_{44}$

$c_{3,2} = -h_{16} - h_{19} - h_{21} - h_{28} - h_{29} - h_{38} + h_{42} + h_{44} - h_{47} + h_{48}$

$c_{4,2} = h_{11} - h_{12} - h_{18} + h_{21} - h_{32} + h_{33} - h_{34} - h_{36} + h_{62} - h_{70}$

$c_{1,3} = h_{15} + h_{23} + h_{24} + h_{34} - h_{37} + h_{40} - h_{41} + h_{55} - h_{56} - h_9$

$c_{2,3} = -h_{10} + h_{19} + h_{32} + h_{35} + h_{36} + h_{37} - h_{43} - h_{60} - h_6 - h_{72}$

$c_{3,3} = -h_{16} - h_{28} + h_{33} + h_{37} - h_{39} + h_{45} - h_{46} + h_{63} - h_{71} - h_8$

$c_{4,3} = h_{10} + h_{15} + h_{16} - h_{33} + h_{34} - h_{35} - h_{37} - h_{54} + h_6 + h_8 - h_9$

$c_{1,4} = -h_{10} + h_{12} + h_{14} - h_{16} + h_{23} + h_{24} + h_{25} + h_{26} + h_5 - h_{66} - h_7$

$c_{2,4} = h_{10} + h_{18} - h_{19} + h_{20} - h_{22} - h_{24} - h_{26} - h_5 - h_{69} + h_{73}$

$c_{3,4} = -h_{14} + h_{16} - h_{23} - h_{26} + h_{27} + h_{29} + h_{31} + h_{46} - h_{58} + h_{76}$

$c_{4,4} = h_{12} + h_{25} + h_{26} - h_{33} - h_{35} - h_{40} + h_{41} + h_{65} - h_{68} - h_7$

$c_{1,5} = h_{15} + h_{24} + h_{25} + h_{27} - h_{28} + h_{30} + h_{31} - h_4 + h_{61} + h_{64}$

$c_{2,5} = -h_{10} - h_{18} - h_2 - h_{30} - h_{38} + h_{42} - h_{43} + h_{46} + h_{67} + h_{74}$

$c_{3,5} = -h_{10} + h_{12} - h_{15} + h_{28} + h_{29} - h_2 - h_{30} - h_3 + h_{46} + h_4 - h_{75}$

$c_{4,5} = -h_{12} - h_{29} + h_{30} - h_{34} + h_{35} + h_{39} + h_3 - h_{45} + h_{57} + h_{59}$

Sources: https://deepmind.google/blog/discovering-novel-algorithms-with-alphatensor/ ; https://www.nature.com/articles/s41586-022-05172-4

Why Isn't RL Everywhere Yet?

The Hard Truth: RL is difficult

- Deep RL is notoriously **unstable**, **sample-inefficient**, and **difficult to debug**.

Current Wisdom: "If you can reframe your problem as supervised learning, do it."

- (+) Imitation is easier & safer than discovery.
- (-) Limited by their training data.
- (-) What happens when we run out of internet data?

RL is painful today, but worth investing.

Why Is RL Difficult?

Task: Drive to NTOU from Academia Sinica.

- I have never run this line before.
  - I don't have such data in my head.
- Here are 4 possible first actions.

What to do now?

Challenge - Comparing Policies Is Hard

Goal: Policy Optimization.

Challenge: To compare Policy A vs. Policy B, we need a reliable, quantifiable **metric** to measure long-term success in a vast, stochastic world.

- E.g., A *stable* and *accurate* value function $V(\pi)$.

Challenge - Designing the Reward Signal is Hard

Goal: Design immediate reward signal for my ultimate goal.

- Which one takes me to the destination? (**Goal**)
- Which one leads to shortest distance? (**Distance**)
- Which one is fastest? (**Time**)
- Which one has most beautiful view on the way? (**Subjective**)
- Which one has better road conditions?

Challenge - Designing the Reward Signal is Hard

Research Questions:

- Sparse Reward Problem
- Reward Hacking Problem
- RLHF: Instead of a definite reward, we only learn the “preference”.
- Inverse RL: Can we learn the hidden reward function from an policy?

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Reward Hacking in Reinforcement Learning

Date: November 28, 2024 | Estimated Reading Time: 37 min | Author: Lilian Weng

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Reward hacking occurs when a reinforcement learning (RL) agent exploits flaws or ambiguities in the reward function to achieve high rewards, without genuinely learning or completing the intended task. Reward hacking exists because RL environments are often imperfect, and it is fundamentally challenging to accurately specify a reward function.

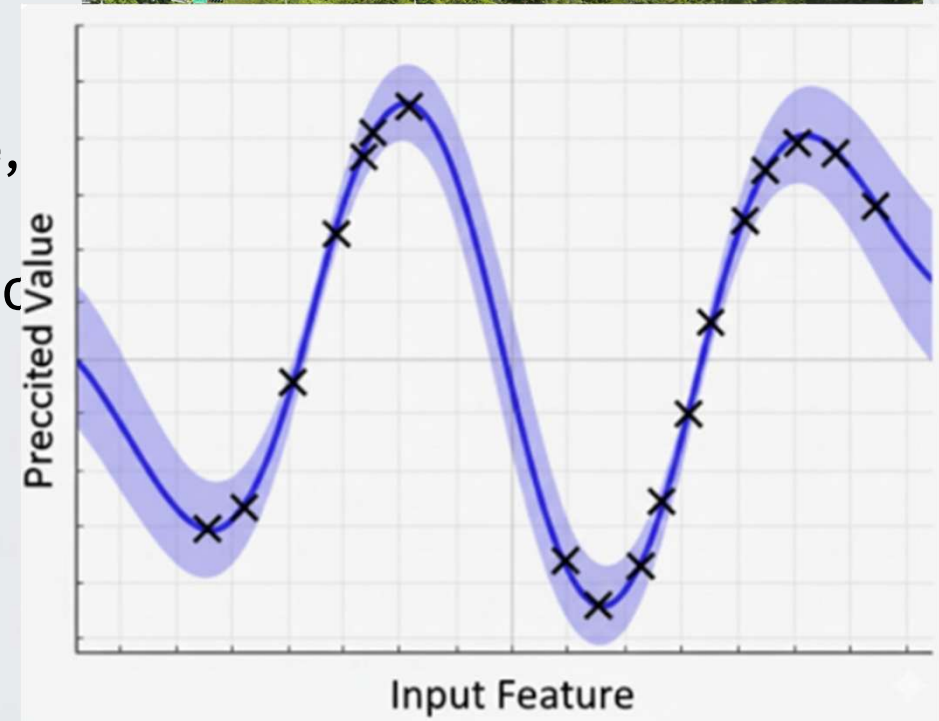
With the rise of language models generalizing to a broad spectrum of tasks and RLHF becomes a de facto method for alignment training, reward hacking in RL training of language models has become a critical practical challenge. Instances where the model learns to modify unit tests to pass coding tasks, or where responses contain biases that mimic a user’s preference, are pretty concerning and are likely one of the major blockers for real-world deployment of more autonomous use cases of AI models.

Most of the past work on this topic has been quite theoretical and focused on defining or demonstrating the existence of reward hacking. However, research into practical mitigations, especially in the context of RLHF and LLMs, remains limited. I especially want to call out for more research efforts directed toward understanding and developing mitigation for reward hacking in the future. Hope I will be able to cover the mitigation part in a dedicated post soon.

Challenge - The Counterfactual Problem

Goal: We must compare **unseen actions**.

- If an action was never tried, we don't know what would have happened (counterfactuals).
- This leads to extremely **high variance**, in our estimates.
- The only way to beat high variance is to collect **massive amounts of data**.



Challenge - Data Is Expensive

Counterfactual & real data are expensive

- We *do have* plenty of **"good" data** (from logs, demos, simulators…) (We usually perform good policies from our "common sense" and "experience".)
- But we lack **"bad" data & counterfactual data**.

Develop ***sample-efficient RL*** to make every bit of experience count.

Key Directions for Sample-Efficient RL

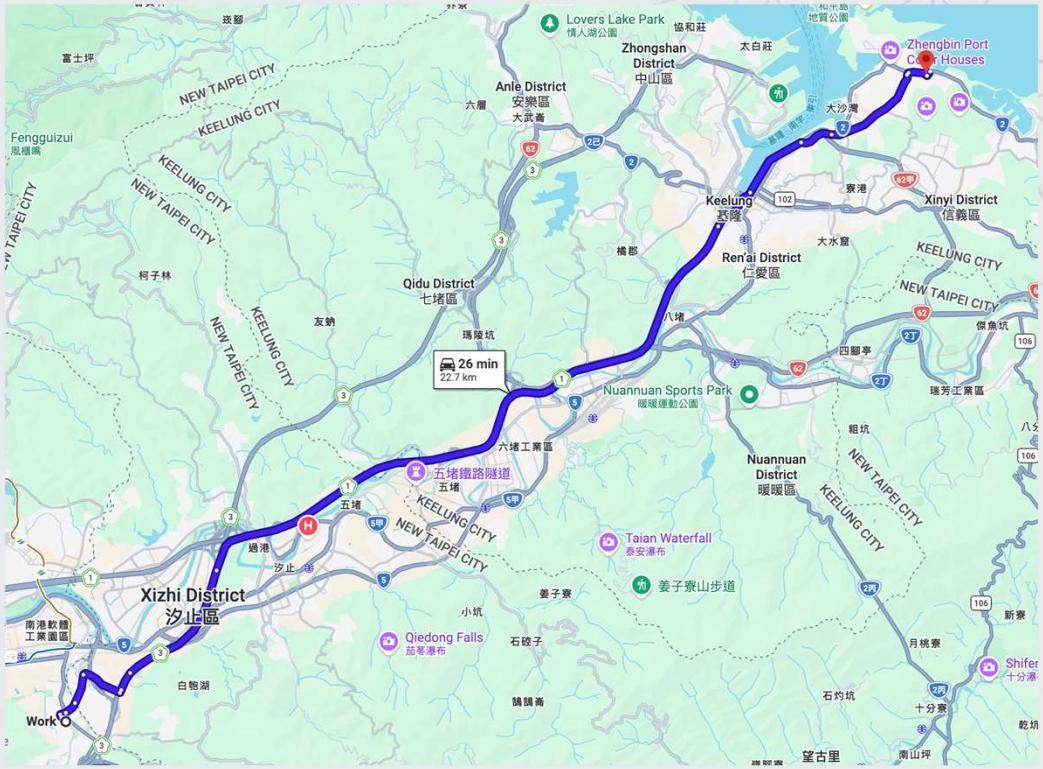
- Squeeze More from Existing Data (Data Reuse)
- Smart Exploration (Data Collection)
- Make Each Update Reliable (Stabilization)
- Transfer What You Already Know (Pretraining/Transfer)

Squeeze More from Existing Data

- To learn a world model (model-based RL)

World Models (Model-Based RL)

- Learn a **predictive model** of the environment.
- Find the desired policy by **planning**.
- Generate **imagined rollouts** internally to create synthetic, cheap data for training.



Squeeze More from Existing Data

- To learn a world model (**Model-Based RL**)
- To learn better features/estimators (**Representation Learning**)
  - Compress high-dimensional inputs into a compact, representative latent.
- To reuse existing data better (**Experience Replay**)
  - Classically: replay buffer is just storage
  - Some modern view: large replay buffer + clever reuse is seen “model-based-ish”
    - Non-parametric approximation of the world
    - Does not really allow you to “predict”

Smart Exploration (Asking the Right Questions)

- **Extrinsic (Goal-Driven) Exploration**
  - Methods like epsilon-greedy or optimism aim to maximize the task return.
- **Intrinsic (Self-Supervised) Exploration**
  - Choose actions that improve knowledge, regardless of task reward.
  - **Common Signals**
    - Novelty/Coverage (states never seen; actions never taken)
    - Prediction Error (where your current model predictor errs most)
    - Disagreement/uncertainty (states where your estimators / predictors disagree)

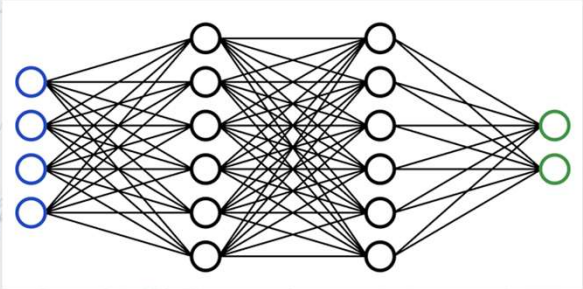
Balance between two explorations is important

Challenge - Value Calculation is Unstable

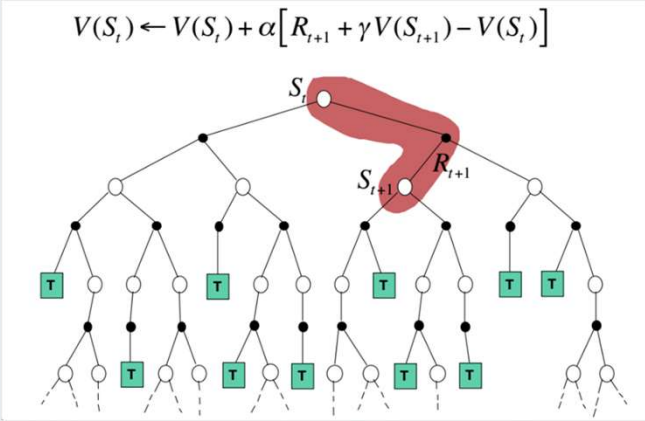
- **High Variance:** need to estimate counterfactual (unobserved) actions.
- **The Deadly Triad**: the risk of divergence arises when these meet.

Function Approximation

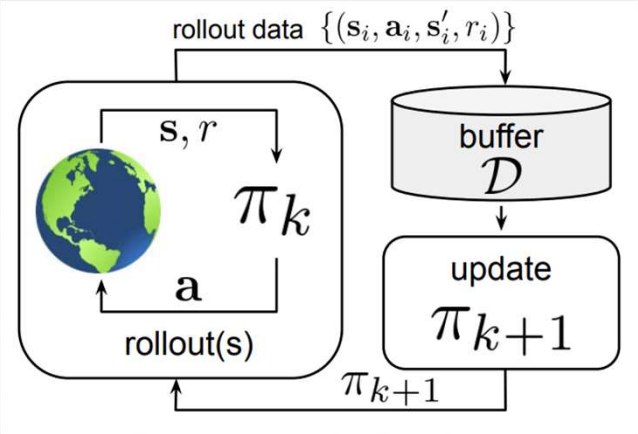
E.g., Neural Network



Bootstrap Update



Off-Policy Learning



Sutton and Barto (2018)
Levine, et al. (2020). Offline reinforcement learning: Tutorial, review, and perspectives on open problems.

Make Each Update Reliable (Stabilization)

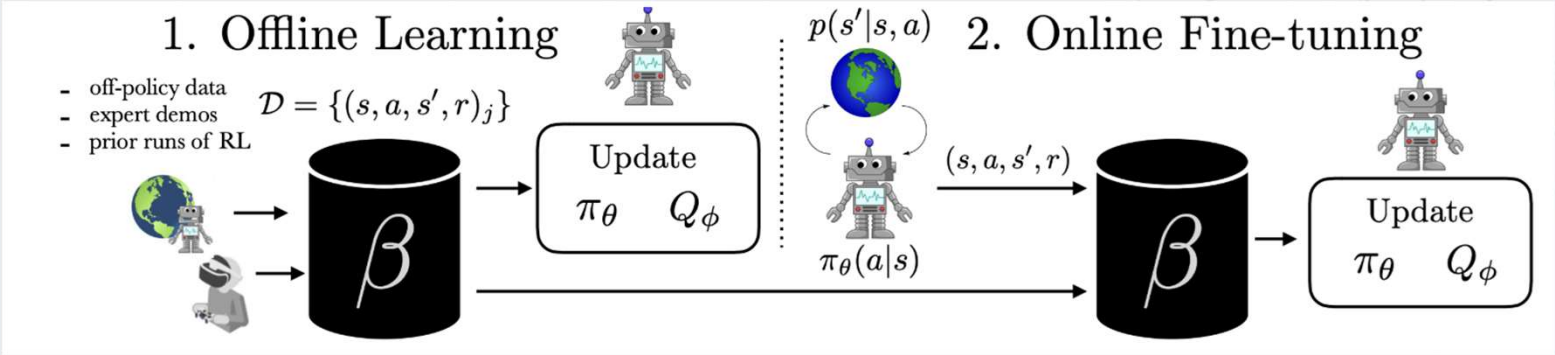
Goal: Reduce bias/variance so every sample pushes the policy/value estimates forward stably.

- **Policy**: Use smooth, constrained policy updates, e.g., TRPO/PPO clipping and entropy bonuses.
- **Value**: Use "conservative updates" to stabilize value updates, e.g., using target network.

Smart Exploration & Reliable Estimate are related.

Transfer What You Already Know (Transfer & Pretrain)

Offline RL



Zero-Shot RL

An agent that can solve any RL task in a given environment, instantly with no additional planning or learning.

Nair and Gupta (2020). AWAC: Accelerating Online Reinforcement Learning with Offline Datasets

Why Is RL Difficult?

The world is large, complex, and full of dynamics.

The goals are hard to define in quantifiable ways.

The "Bitter Lesson" by Richard Sutton (2019)

AI researchers have often tried to build knowledge into their agents.

- helps in the short term, and is personally satisfying to the researcher.
- but in the long run it plateaus and even inhibits further progress
- and breakthrough progress eventually arrives by an opposing approach based on scaling computation by search and learning.

"Time spent on one is time not spent on the other."

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Thank You